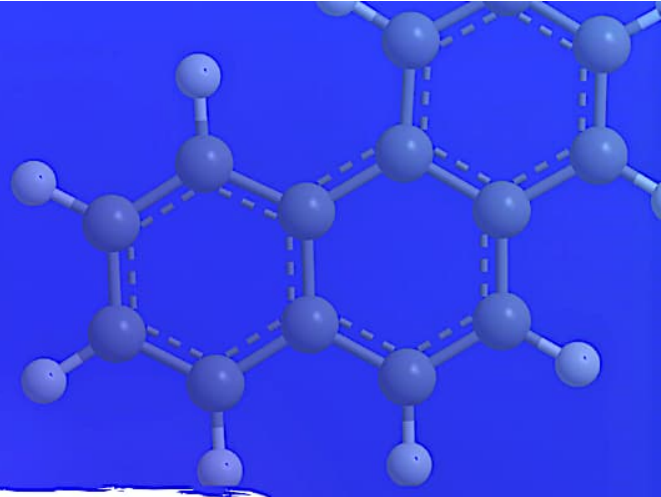




Class 9

Chapter 3

Dynamics

Chapter 3 | Class 9 | Physics

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Chapter

3

Dynamics

3.1

Concept of Force

Multiple Choice Questions (MCQs)

1. Friction is a:
(A) Contact force (B) Non-contact force
(C) Nuclear force (D) Magnetic force
2. The force acting along a rope is:
(A) Thrust force (B) Tension force
(C) Non-contact force (D) Drag
3. The value of 'G' is:
(A) $6.67 \times 10^{-11} \text{ Nm}^2 \text{ Kg}^{-2}$ (B) $6.67 \times 10^{-9} \text{ Nm}^2 \text{ Kg}^2$
(C) $6 \times 10^{-11} \text{ Nm}^2 \text{ Kg}^{-2}$ (D) $6.67 \times 10^{24} \text{ Nm}^2 \text{ Kg}^{-2}$
4. The force between sub-atomic particles is:
(A) Electrostatic force (B) Nuclear force
(C) Normal force (D) Contact force
5. Upward force exerted by a liquid is called:
(A) Drag (B) Thrust (C) Normal force (D) Friction
6. Non-contact force is also called:
(A) Friction (B) Thrust (C) Field force (D) Drag

Short Questions

1. Define forces. Give examples?

Ans. Force: A force is a push or a pull that starts, stops or changes the magnitude and direction of velocity of a body.

Example: (i) When we open a door, we push or pull it by applying force.

- (ii) When we are sitting in a car, we push against the seat as the car turns round a corner.

2. Define contact force. Give some examples.

Ans. Contact Forces: A contact force is a force that is exerted by one object on the other at the point of contact.

Examples: (i) Friction (ii) Drag (iii) Thrust

3. Define friction and drag force.

Ans. (i) Friction: It is the force that resists motion when the surface of one object comes in contact with the surface of another.

(ii) Drag: The drag force is the resistant force caused by the motion of a body through a fluid. It acts opposite to the relative motion of any object moving with respect to surrounding fluid.

4. Define thrust and normal force.

Ans. (i) Thrust: It is an upward force exerted by a liquid on an object immersed in it.

When we try to immerse an object in water, we feel an upward force exerted on the object. A ship can float in the sea due to this force which balances the weight of the ship.

(ii) Normal Force: It is the force of reaction exerted by the surface on an object lying on it. This force acts outward and perpendicular to the surface. It is also called the support force upon the object, e.g. the force exerted by the table on the book lying on it, is a normal force.

5. Define air resistance, tension force and elastic force.

Ans. (i) Air Resistance: It is the resistance (opposition) offered by air when an object falls through it.

(ii) Tension Force: It is the force experienced by a rope when a person or load pulls it.

(iii) Elastic Force: It is a force that brings certain materials back to their original shape after being deformed.

Examples: Rubber bands, springs, trampoline, etc.

6. What are non-contact forces? Give examples.

Ans. Non-contact force: A non-contact force is defined as the force between two objects which are not in physical contact. The non-contact forces can work from a distance. That is why, these are sometimes called as action-at-a-distance.

Example: (i) Gravitational Force (ii) Electrostatic Force

(iii) Magnetic Force (iv) Strong and Weak Nuclear Forces

7. What do you mean by gravitational force? Write its formula?

Ans. Gravitational Force: The gravitational force is an attractive force that exists among all bodies which have mass. It is a long-range force given by Newton's law of gravitation:

Formula:

$F = G \frac{m_1 m_2}{r^2}$ where m_1 and m_2 are two masses distant r apart and G is constant of gravitation. Its value is $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

8. What do you mean by electrostatic force and strong and weak nuclear force.

Ans. Electrostatic Force: An electrostatic force is a force that acts between two charged objects. The opposite charges attract each other and similar charges repel each other.

Strong and Weak Nuclear Forces: These are non-contact forces acting between the subatomic particles.

9. What do you mean by magnetic force?

Ans. Magnetic Force: It is a force which a magnet exerts on other magnets and magnetic materials like iron, nickel and cobalt. Magnetic force between the poles of two magnets can be either attractive or repulsive. Like poles repel and unlike poles attract each other.

3.2

Fundamental Forces

Multiple Choice Questions (MCQs)

7. The number of fundamental forces in nature:
 (A) 2 (B) 3 (C) 4 (D) 5
8. The force responsible for disintegration of nucleus is:
 (A) Strong nuclear force (B) Weak nuclear force
 (C) Gravitational force (D) Normal force
9. The range of strong nuclear force is:
 (A) 10^{-14} m (B) 10^{-16} m (C) 10^{-6} m (D) 10^{-7} m
10. Unchange particle in β -decay is:
 (A) Electron (B) Proton (C) Neutron (D) Antineutrino
11. Dr. Abdus Salam got Noble prize in:
 (A) 1979 (B) 1989 (C) 1879 (D) 1779
12. The range of weak nuclear force is:
 (A) 10^{-17} m (B) 10^{-11} m (C) 10^{-7} m (D) 10^{-18} m

Short Questions

1. Write the names of fundamental forces?

Ans. **Fundamental Forces:** There are four fundamental forces in nature.

- | | |
|-------------------------|--------------------------|
| 1. Gravitational force | 2. Electromagnetic force |
| 3. Strong nuclear force | 4. Weak nuclear force |

2. What do you mean by gravitational force?

Ans. Gravitational force is an attractive force that exists among all bodies which have mass.

3. What do you mean by electromagnetic force?

Ans. Electromagnetic force is the force that causes the interaction between electrically charged particles. The area in which these forces act are called electromagnetic field. It binds together atoms, molecules and crystals.

4. What are strong nuclear forces? Explain.

Ans. Strong nuclear force holds the atomic nuclei together by binding the protons and neutrons in the nucleus over-coming repulsive electromagnetic force between positively charged protons. It is also a short-range force with the order of 10^{-14} m. If the distance between nucleons increases beyond this range, this force ceases to act.

5. What are weak nuclear forces? Explain.

Ans. Weak nuclear force is a force responsible for the disintegration of a nucleus. For example, the weak nuclear force executes the β -decay (beta decay) of a neutron, in which a neutron transforms into a proton. In the process, a β -particle (electron) and an uncharged particle called antineutrino are emitted. Due to weak nuclear force radioactive decay of atoms occurs. It is a short-range force of the order 10^{-17} m.

6. Explain the unification of weak nuclear and electromagnetic forces?

Ans. A Pakistani scientist Dr. Abdus Salam, Sheldon Glashow and Steven Weinberg were awarded in 1979 Nobel Prize in Physics for their contributions to unification theory.

They explain the behaviour of weak nuclear force and electromagnetic force is different for everyday phenomena but at very high energy the theory models these forces as a single force called electroweak force. This is called unification of nuclear force and electromagnetic force.

3.3

Forces in a Free- Body Diagram

Multiple Choice Questions (MCQs)

13. External force acts on body:

- (A) Friction (B) Gravity (C) Drag (D) All

14. Free body diagram is example of:

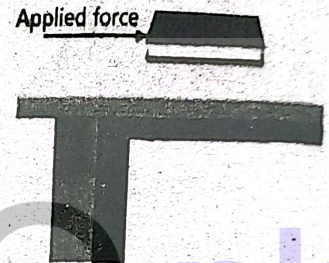
- (A) Simple diagram (B) Vector diagram
(C) Scalar diagram (D) Complex diagram

Short Questions

1. How many external forces act on body? Give examples.

Ans. External forces acting on an object may include:

- (i) friction (ii) gravity
(iii) normal force (iv) drag
(v) tension in a string
(vi) a human force due to pushing or pulling



3.4

Newton's Laws of Motion

Multiple Choice Questions (MCQs)

15. Sir Isac Newton was born in:

- (A) Lincolnshire (B) Africa (C) America (D) Ireland

16. The book principia Mathematical was written by:

- (A) Newton (B) Fleming (C) Faraday (D) Ohm

17. Inertia depends on:

- (A) Force (B) Weight (C) Mass (D) Speed

18. Newton's first law of motion is also called law of:

- (A) Mass action (B) Inertia
(C) Conservation of energy (D) Conservation of mass

19. One newton is equal to:

- (A) 10 kgms^{-2} (B) 1 kgms^{-2} (C) 5 kgms^{-2} (D) 100 kgms^{-2}

20. The S.I unit of force is:

- (A) N (B) Ns (C) Nm (D) N kg^{-1}

21. The gun recoils after firing due to:

- (A) Applied force (B) Drag (C) Thrust (D) Reaction

Short Questions

1. State Newton's first law of motion. Why it is called law of inertia?

Ans. **Statement:** "A body continues its state of rest or of uniform motion in a straight line unless acted upon by some external force."

Law of inertia: As Newton's first law of motion deals with the inertial properties of matter so it is called law of inertia.

2. Define force in term of Newton's first law of motion.

Ans. According to Newton's first law of motion, a force can be defined as:

Force is an agency which changes or tends to change the state of rest or of uniform motion of a body.

3. Define inertia. On what factors does it depend?

Ans. **Inertia:** The property of a body to maintain its state of rest or of uniform motion in a straight line is called inertia.

Factor: It depends on mass. The greater is the mass of an object, the greater is its inertia.

4. When the table cloth is pulled abruptly, the objects remain in their original position on the table. Why?

Ans. When the table cloth is pulled abruptly, the objects remain in their original position on the table due to inertia which maintains its state of rest.

5. State Newton's second law of motion?

Ans. **Statement:** "If a net external force acts upon a body, it accelerates the body in the direction of force. The magnitude of acceleration is directly proportional to the magnitude of force and is inversely proportional to the mass of the body."

6. Write mathematical form of Newton's second law of motion?

Ans. **Mathematical form:** The second law can be written mathematically as:

$$a \propto F$$

$$a \propto \frac{1}{m}$$

$$a \propto \frac{F}{m}$$

$$a = (\text{constant}) \frac{F}{m}$$

According to SI units, if $m = 1 \text{ kg}$, $a = 1 \text{ ms}^{-2}$, $F = 1 \text{ N}$, then the value of the constant will be 1. Therefore, the above equation can be written as:

$$a = 1 \times \frac{F}{m}$$

or $F = ma$

7. Define 1 Newton force.

Ans. Newton: One newton is the force which produces an acceleration of 1 ms^{-2} in a body of mass 1 kg .

$$1 \text{ N} = 1 \text{ kg ms}^{-2}$$

8. State Newton's third law of motion with examples?

Ans. Statement: For every action, there is always an equal and opposite reaction.

Examples:

- (i) Consider a block lying on a table. The block exerts a downward force on the table equal to its weight w . The table also exerts a reaction force F , on the block.
- (ii) When a bullet is fired from a gun, the bullet moves in the forward direction with a force F . This is the force of action. The gun recoils in the backward direction with a reaction force R .

3.5 Limitations of Newton's Laws of Motion

Multiple Choice Questions (MCQs)

22. Newton's laws of motion are not applicable for:

- | | |
|-------------------|--------------------------|
| (A) Large object | (B) Medium size objects |
| (C) Small objects | (D) Elementary particles |

23. Relativistic mechanics was developed by:

- | | | | |
|------------|-------------|--------------|-----------|
| (A) Newton | (B) Fleming | (C) Einstein | (D) Henry |
|------------|-------------|--------------|-----------|

Short Questions

1. What are the limitations of Newton's laws of motion?

Ans. Newton's laws of motion are not exact for all types of motion, but provide a good approximation, unless an object is small enough or moving close to the speed of light. There are limitations of Newton's laws of motion when we deal with the motion of elementary particles having velocities close to that of light. For that purpose, relativistic mechanics developed by Albert Einstein is applicable.

3.6

Mass and Weight

Multiple Choice Questions (MCQs)

24. The force with which earth attracts the body is called:
 (A) Contact force (B) Drag force
 (C) Resistance (D) Weight
25. Gravitational field strength near the surface of earth is:
 (A) 10 N kg^{-1} (B) 20 N kg^{-1} (C) 30 N kg^{-1} (D) 40 N kg^{-1}
26. Weight is measured by:
 (A) Ordinary balance (B) Physical balance
 (C) Lever balance (D) Spring balance
27. SI unit of weight is:
 (A) Kg (B) N (C) KN (D) g

Short Questions

1. Differentiate between mass and weight.

Ans. Difference between mass and weight:

Mass	Weight
(i) The amount of matter in a substance is called the mass.	(i) The weight of a body on the earth is the force with which the earth pulls the body towards itself.
(ii) It is a scalar quantity.	(ii) It is a vector quantity.
(iii) The amount of mass does not change when a body is moved from one place to another.	(iii) Weight depends on gravitational acceleration (g) and its magnitude changes with change of position.
(iv) Its unit is kg.	(iv) Its unit is N.
(v) It is measured by physical balance.	(v) It is measured by spring balance.

2. Define gravitational field.

Ans. The gravitational field is a space around a mass in which another mass experiences a force due to gravitational attraction.

3. What is gravitational field strength? Write its value?

Ans. Gravitational field strength: The gravitational field strength is defined as the gravitational force acting on unit mass.

Value: Its value is 10 N kg^{-1} .

4. Why does weight of a body varies from place to place?

Ans. The weight of a body depends upon the value of g . As the value of g varies from place to place, therefore, the value of weight does not remain the same everywhere.

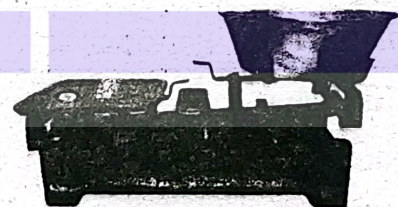
3.7**Mechanical and Electronic Balances****Multiple Choice Questions (MCQs)**

28. Which instrument is called Newton meter:
Ⓐ Force meter Ⓑ Lactometer Ⓒ Ammeter Ⓓ Manometer
29. More precise balance is:
Ⓐ Electronic balance Ⓑ Physical balance
Ⓒ Beam balance Ⓓ Lever balance
30. Standard weights are not required for:
Ⓐ Beam balance Ⓑ Lever balance
Ⓒ Electronic balance Ⓓ Physical balance
31. A digital force meter measures directly the weight in:
Ⓐ g Ⓑ Kg Ⓒ N Ⓓ Nm^{-2}
32. The weight of 100 g is equal to:
Ⓐ 1N Ⓑ 3N Ⓒ 2N Ⓓ 4N

Short Questions

1. Write down the construction and working of mechanical balance?

Ans. **Construction:** A mechanical balance consists of a rigid horizontal beam that oscillates on a central knife edge as a fulcrum.



Working: It has two end knife edges equidistant from the centre. Two pans are hung from bearings on the end knife edges. The material to be weighed is put in one pan. Standard weights are put on the other pan. Add standard weights until pointer attached to the beam comes in equilibrium. Now, total standard weight is the weight of the unknown object.

2. What are balance scales?

Ans. The scales which are used to measure mass or weight of an object by comparing it with standard weights is called balance scale.

3. Define electronic balance. Write its working.

Ans. **Electronic balance:** Electronic balance is used in laboratory to measure the mass of different objects.

Working: No standard weights are required to use in an electronic balance. Only it has to be connected to a power supply. When an object is placed on it, its mass is displayed on its screen. Electronic balances also display the total price of the material if the rate per kg is fed to the balance.

4. **What is a force meter?**

Ans. Force meter: A force meter is a scientific instrument that measures force. It is also called as a newton meter or a spring balance. Now a days digital force meters are also available. Force meter measures force directly in newtons (N).



5. **How weight of an object is measured with the help of force meter?**

Ans. The object to be weighed is hung with the hook. The mass of the object causes the spring to compress. The pointer indicates the weight of the object.

3.8

Friction**Multiple Choice Questions (MCQs)**

33. The force which opposes the motion is called:
(A) Nuclear force (B) Non-contact force
(C) Restoring force (D) Friction
34. Which type of force is a friction:
(A) Dissipative force (B) Applied force
(C) Basic force (D) Thrust
35. The force of friction after motion is:
(A) Static friction (B) Kinetic friction
(C) Rolling friction (D) All of these
36. The surface temperature when a shuttle re-enters the earth atmosphere is:
(A) 900°C (B) 950°C (C) 190°C (D) 450°C
37. Which friction is the smallest:
(A) Sliding friction (B) Static friction
(C) Kinetic friction (D) Rolling friction
38. Friction depends on:
(A) Area of surface (B) Nature of surface
(C) Speed of body (D) Physical conditions

Short Questions

1. Define friction. Explain with example?

Ans. Friction: Friction is the force that tends to prevent the bodies from sliding over each other.

Explanation: When a cricket ball is hit by the bat, it moves on the ground with a reasonably large velocity. According to Newton's first law of motion, it should continue to move with constant velocity. But, practically, we observe that it eventually stops after covering some distance. It stops due to friction.

2. How energy is dissipated during friction?

Ans. Dissipation of energy: Friction is a dissipative force due to which the energy is wasted in doing work to overcome against friction. The lost energy appears in the form of heat.

Example: When we rub our hands, heat is produced due to friction and our hands become warm.

3. Define sliding friction. Write its types?

Ans. Sliding friction: The friction between two solid surfaces is called sliding friction.

Types:

(i) Static friction (ii) Kinetic friction

4. Differentiate between static and dynamic friction.

Static Friction	Kinetic Friction
Static friction is the force of friction arising due to applied external force before motion of one body over the other.	Kinetic friction is the force of friction arising due to applied external force after motion of one body over the other.

5. Define terminal velocity. Give example?

Ans. Terminal Velocity: When upward air resistance balances the downward force of gravity on a falling object, it falls down with constant (safe) velocity, it is called terminal velocity.

Example: A paratrooper coming down with terminal velocity.

6. What is rolling friction? How rolling friction is lesser than sliding friction?

Ans. Rolling friction: When an object rolls over a surface, the friction produced is called rolling friction.

Reason: The reason for the rolling friction to be less than the sliding friction is that there is no relative motion between the wheel and the surface over which it rolls. Moreover the wheel touches the surface only at one point which reduces the sliding friction.

7. Write down methods of reducing friction?

Ans. The following methods are used to reduce friction:

- (i) The parts which slide against each other are highly polished to reduce friction.
- (ii) Oil or grease is applied between the moving parts of the machinery to reduce friction.
- (iii) As rolling friction is much less than the sliding friction, so sliding friction is converted into rolling friction by the use of ball bearings in the machines and wheels under the heavy objects.

8. Why bodies of moving vehicles are streamlined?

Ans. If the front of a vehicle is flat, it faces more resistance by air or water. Therefore, the bodies moving through air or water are streamlined to minimize air or water friction. In this case, the air passes smoothly over the slanting surface of vehicle. This type of flow of air is known as streamline flow.

9. Why frogs can cling over vertical surface?

Ans. Some frogs can cling to a vertical surface, such as leaf, because of the static friction between their feet and the surface.

10. What purpose do the spaces between tread pattern on tyre serve?

Ans. On a wet road, the water does not form wet layer between the tyre surface and the road surface due to the spaces in the tread pattern on the tyre. This reduces the chances of skidding of vehicles on wet roads.

11. Why our bones do not normally wear out after many years?

Ans. Friction in human joints is very low because our bodies contain a natural lubricating system. Consequently, though our bones rub against each other at the points as we move, yet bones do not normally wear out, even after many years of use.

12. What is hovercraft? How does it move?

Ans. Hovercraft: A hovercraft is a kind of ship that can move over the surface of water and ground both.

Working: Air is ejected underneath the hovercraft by powerful fans forming a cushion of air. The hovercraft moves over the cushion of air which offers very small resistance.

13. What purpose does the flat portion of a wheel serve?

Ans. The flat portion of the wheel has tendency to slide against the surface and does produce a frictional force. This enables the vehicle to move smoothly.

3.9**Momentum and Impulse****Multiple Choice Questions (MCQs)**

39. The product $F \cdot \Delta t$ is called:

- (A) Momentum (B) Impulse (C) Inertia (D) Elasticity

40. Rate of change of momentum is equal to:

- (A) Inertia (B) Acceleration (C) Pressure (D) Force

41. Unit of momentum is:

- (A) Kg ms^{-2} (B) N Kg^{-1} (C) Ns (D) Nm

Short Questions

1. Why fragile objects are packed in soft materials like styrofoam boxes?

Ans. Fragile objects such as glassware may break easily due to jerks or by the direct impact with hard objects during their transportation.

To protect them soft, packing materials are used for these objects. These materials reduce the effect of quick change in momentum. Consequently, the force acting on the fragile objects is greatly reduced.

2. What is a crumple zone?

Ans. A crumple zone of an automobile is a structural feature designed to compress during an accident to absorb deformation energy from the impact.

3. Define momentum. On what factors does it depend?

Ans. **Momentum:** The product of mass and velocity of a moving body is called momentum.

Formula: $p = m \times v$

Unit: The SI unit of momentum is (kg ms^{-1}) . It can also be written as (Ns).

Factors: It depends on mass and velocity of an object. Greater is the mass, the greater will be the quantity of motion. Similarly, greater is the velocity, the greater will be quantity of motion.

4. **Why crumple zones are made in front and behind the main body of the vehicles?**

Ans. Typically, crumple zones are located in front and behind the main body of the vehicle because they are designed to compress during an accident to absorb deformation energy from the impact.

Crumple zones work by managing crash energy absorbing within the outer parts of the vehicle, rather than being directly transmitted to the occupants.

3.10 Principle of Conservation of Momentum

Multiple Choice Questions (MCQs)

42. **Collection of objects is called:**
 (A) Molecule (B) System (C) Ideas system (D) Compound
43. **Principle of conservation of momentum is applicable for:**
 (A) Macro objects (B) Atom (C) Molecules (D) All of these

Short Questions

1. **Define impulse.**

Ans. Impulse: It is the product of force acting on a body for a very short interval of time t .

Formula:

$$\text{Impulse} = F \times \Delta t$$

Unit: Its unit is Ns.

2. **State principle of conservation of momentum.**

Ans. Statement: If no external force acts on an isolated system, the final total momentum of the system is equal to the initial total momentum of the system.

Equation:

$$\begin{array}{lcl} \text{Total momentum of the} & = & \text{Total momentum of the} \\ \text{system before collision} & & \text{system after collision} \\ \text{or } m_1 v_1 + m_2 v_2 & = & m_1 v'_1 + m_2 v'_2 \end{array}$$

3. What is purpose of seatbelt in vehicles?

Ans. When a moving car stops suddenly, the passengers move forward toward the windshield. Seatbelts prevent the passengers from moving. Thus, chances of hitting the passengers against the windshield or steering wheel are reduced.

MCQ's Key

1. (A)	2. (B)	3. (A)	4. (B)	5. (B)	6. (C)	7. (C)	8. (B)	9. (A)	10. (D)
11. (A)	12. (A)	13. (D)	14. (B)	15. (A)	16. (A)	17. (C)	18. (B)	19. (B)	20. (A)
21. (D)	22. (D)	23. (C)	24. (D)	25. (A)	26. (D)	27. (B)	28. (A)	29. (A)	30. (C)
31. (C)	32. (A)	33. (D)	34. (A)	35. (B)	36. (B)	37. (D)	38. (B)	39. (B)	40. (D)
41. (C)	42. (B)	43. (D)							

EXERCISE

A. Multiple Choice Questions

3.1. When we kick a stone, we get hurt. This is due to:

- (A) inertia (B) velocity (C) momentum (D) reaction

3.2. An object will continue its motion with constant acceleration until:

- (A) the resultant force on it begins to decrease.
 (B) the resultant force on it is zero.
 (C) the resultant force on it begins to increase.
 (D) the resultant force is at right angle to its tangential velocity.

3.3. Which of the following is a non-contact force?

- (A) Friction (B) Air resistance
 (C) Electrostatic force (D) Tension in the string

3.4. A ball with initial momentum p hits a solid wall and bounces back with the same velocity. Its momentum p after collision will be:

- (A) $p' = p$ (B) $p' = -p$ (C) $p' = 2p$ (D) $p' = -2p$

3.5. A particle of mass m moving with a velocity v collides with another particle of the same mass at rest. The velocity of the first particle after collision is:

- (A) v (B) $-v$ (C) 0 (D) $-1/2$